

Gholibjon Qasobov

Automation and Robotics Engineer

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Skills and Interests

Programming Languages: C++, Python, MATLAB, IEC 61131-3 (ST, LD, FBD)

Robotics: ROS1 Noetic, ROS2 Humble, RoboDK, Nav2, Gazebo, Computer Vision, Machine Learning, Reinforcement Learning, NVIDIA Isaac Sim & Isaac Lab

Electronics and Hardware Skills: Raspberry Pi 4/5, ESP32, Nvidia Jetson, Modicon M340/M241 PLCs, Lidar (LS-LIDAR-C32, LDS-02), Intel RealSense D435i

Design & Prototyping: Autodesk Fusion 360, 3D Printing

Software & Tools: Git, GitHub, Linux, Docker, Jenkins

Experience

Robotics & RL Engineer, Omnibio, England, UK (remote) January 2026 - Present

- Implemented parallel **MPPI-based trajectory optimization** using Active Inference and **NVIDIA Isaac Lab** for robust control under uncertainty.
- Integrated **Active Inference** pipelines (Julia) with ROS2 and **NVIDIA Isaac Sim agents**, enabling closed-loop perception-action loops.
- Designed and developed simulation environments for industrial manipulators (**Fanuc, UR, KUKA**) in **NVIDIA Isaac Sim**, including conveyors, object flows, and task-specific end-effectors.
- Built modular simulation pipelines for rapid prototyping and validation of manipulation tasks in digital twin environments.

Robotics Intern (short term), Arcobo, Almaty, Kazakhstan April 2026

- Deployed and operated a **Kawasaki BA006L** industrial robot
- Programmed and executed motion tasks on the **Astorino Kawasaki** platform, including trajectory following and pick-and-place operations.
- Utilized **SimBox HIL simulation** for safe testing, validation, and rapid iteration of robotic workflows.

Robotics Laboratory Assistant, Kazakh-British Technical University January 2024 - Dec 2025

- Developed an **automated inspection system** using a **mobile-legged robot** with integrated elevator control, reaching **95 % inspection accuracy** and **90 % success in multi-floor transitions**.
- Collaborated on designing and simulating an innovative dual-function end-effector integrating a screwdriver and suction cup for **automated power bank assembly using RoboDK** ([JMST 2025 Conference paper, under review after revision](#))
- Built an **AprilTag**-based pick-and-place system using a **DoBot** robotic arm, enabling **95 % positioning accuracy** and autonomous object handling within a 30 cm workspace.

Robotics Software Engineer, Alma Valley - Borealis Laboratory November 2024 – May 2025

- Designed an autonomous water drone for environmental cleanup using **NVIDIA Jetson Orin NX** and **RealSense D435i**, achieving **90 % detection accuracy** and 2 kg debris capacity.
- Developed a **Kazakh voice control system** for the **Unitree Go2** robot using an **offline LLM** integrated with ROS2 Humble.

Research Intern, Constructor Knowledge Labs, Zurich, Switzerland (remote) June 2025 - August 2025

- Researched and implemented **Retrieval-Augmented Generation (RAG)** for multi-domain question answering.
- Focused on optimizing context retrieval and structured response generation using **Large Language Models (LLMs)**.

- Evaluated system performance across diverse domain corpora to improve accuracy and coherence.

Selected Projects

Vision-Based Teleoperation of Unitree G1 Humanoid Robot

- Developed a **vision-based teleoperation** system using MediaPipe for **real-time human pose estimation**.
- Implemented **inverse kinematics** and **collision-aware retargeting** for safe humanoid arm motion.
- Integrated full pipeline in **ROS2** and validated in **NVIDIA Isaac Sim** before deployment on real robot.
- Tools Used: Python, C++, ROS2 Humble, MediaPipe, Isaac Sim, Rviz

Reinforcement Learning-Based Locomotion for Unitree Go

- Designed and trained a **PPO-based locomotion policy** in **Isaac Lab** for velocity tracking and stable gait generation.
- Formulated MDP, including observation/action spaces and reward functions for stability and efficiency.
- Deployed and validated trained policy in **NVIDIA Isaac Sim**.
- Tools Used: Python, ROS2 Humble, Isaac Lab, Isaac Sim

Autonomous Warehouse Mobile Robot Integration Using TurtleBot3 and OPC U

([Publication](#) )

- Collaborated on developing a LIDAR-based navigation and **grid-positioning algorithm** for the TurtleBot3 Waffle Pi to enable autonomous warehouse operation.
- Integrated a 3D-printed lifting mechanism and **OPC UA connectivity with Wiren Board 7 and Honeywell Experion PKS** for seamless industrial communication.
- Tools Used: Python, ROS1 Noetic, Wiren Board, Honeywell Experion PSK, Fusion360

Autonomous Multi-Floor Inspection Mobile Legged-Robot with Elevator

Integration ([Nature Scientific Journal \(Q1\)](#), [under review after revision](#))

- Designed and implemented an autonomous inspection mobile legged-robot capable of elevator-assisted multi-floor navigation, achieving **95 % inspection precision** and **90% reliability in floor transitions**.
- Tools Used: C++, Python, ROS2 Humble, Nav2, Fusion-360, OpenCV, YOLO

Activities and Achievements

Digital Bridge 2025 Robotics Hackathon Winner: Developed an autonomous navigation system for the Unitree Go2 robot dog to patrol complex maze-like environments, with a custom web interface for setting routes and viewing live camera feedback.

KazEnergy Eco-Shell Finalist: Recognized for the innovative design and implementation of a Semi-Autonomous Water Drone for trash collection.

Electrical Engineering Olympiad Finalist: Demonstrated excellence in problem-solving and electrical engineering concepts at olympiad held by Satpayev University.

Judge in Robo-Football competition: Served as a judge in Robo-Football competition in ITFest 2024 & 2025.

Education

Kazakh-British Technical University, BS in Engineering and Technology

Sept 2022 – May 2026

- **Coursework:** Foundations of Electrical Engineering, Theory of Linear and Non-linear Control Systems, Autonomous Mobile Robots, Industrial Robot Operations, Introduction to Machine Learning, PLC Programming, Automation Components and Devices, Robotics in Manufacturing

Portfolio

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